

Matrices (AI HL)

ANSWER KEY



Matrix arithmetic

- 1 Matrix A has rows $(2, -1)$ and $(3, 4)$; matrix B has rows $(0, 5)$ and $(-2, 1)$. Find the rows of $A + 2B$.
 $2B$ has rows $(0, 10)$ and $(-4, 2)$. Adding row by row: $A + 2B$ has first row $(2 + 0, -1 + 10) = (2, 9)$ and second row $(3 - 4, 4 + 2) = (-1, 6)$.
- 2 For the same matrices A (rows $(2, -1), (3, 4)$) and B (rows $(0, 5), (-2, 1)$), find the rows of the product AB , using row-times-column multiplication.
Entry (row 1, col 1): $2(0) + (-1)(-2) = 2$. Entry (row 1, col 2): $2(5) + (-1)(1) = 9$. Entry (row 2, col 1): $3(0) + 4(-2) = -8$. Entry (row 2, col 2): $3(5) + 4(1) = 19$. So AB has first row $(2, 9)$ and second row $(-8, 19)$.

Determinants and inverses of 2×2 matrices

- 3 Matrix M has rows $(3, 5)$ and $(2, 4)$. Find $\det(M)$, using $\det(M) = ad - bc$ for rows (a, b) and (c, d) , and use it to determine whether M is invertible.
 $\det(M) = 3(4) - 5(2) = 12 - 10 = 2$. Since $\det(M) \neq 0$, M is invertible.
- 4 Find the inverse of M (rows $(3, 5)$ and $(2, 4)$), using $M^{-1} = \frac{1}{\det(M)}$ times the matrix with rows $(d, -b)$ and $(-c, a)$.
From the previous problem, $\det(M) = 2$. $M^{-1} = \frac{1}{2} \times$ the matrix with rows $(4, -5)$ and $(-2, 3)$, giving M^{-1} with rows $(2, -2.5)$ and $(-1, 1.5)$.

Solving systems of equations using matrices

- 5 Write the system $3x + 5y = 11$, $2x + 4y = 8$ in matrix form, then use M^{-1} (found in the previous problem, rows $(2, -2.5)$ and $(-1, 1.5)$) to solve for x and y .
Matrix form: $M(x, y)^T = (11, 8)^T$, where M has rows $(3, 5)$ and $(2, 4)$ (i.e. the system's coefficient matrix acting on the column vector of unknowns equals the column vector of constants). Applying M^{-1} :
 $x = 2(11) + (-2.5)(8) = 22 - 20 = 2$. $y = (-1)(11) + 1.5(8) = -11 + 12 = 1$. So $x = 2$, $y = 1$.

Eigenvalues and eigenvectors

- 6 Matrix M has rows $(4, 1)$ and $(2, 3)$. Find the eigenvalues of M , using $\det(M - \lambda I) = 0$.
 $M - \lambda I$ has rows $(4 - \lambda, 1)$ and $(2, 3 - \lambda)$.
 $\det = (4 - \lambda)(3 - \lambda) - 2 = 12 - 7\lambda + \lambda^2 - 2 = \lambda^2 - 7\lambda + 10 = 0$. Factoring: $(\lambda - 5)(\lambda - 2) = 0$, so $\lambda = 5$ or $\lambda = 2$.

- 7 Find the eigenvector corresponding to $\lambda = 5$ for the matrix M (rows (4, 1) and (2, 3)) from the previous problem.
- Solve $(M - 5I)\mathbf{v} = \mathbf{0}$, where $M - 5I$ has rows $(-1, 1)$ and $(2, -2)$. From the first row: $-x + y = 0$, so $y = x$. Eigenvector: $(1, 1)$ (or any scalar multiple).
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Applying matrices to transition/Markov models

- 8 A weather model has a transition matrix T with first row (0.7, 0.4) and second row (0.3, 0.6) (columns: today's sunny/rainy state; rows: tomorrow's sunny/rainy probabilities). If today is sunny (state vector with entries 1 then 0), find the probability distribution for tomorrow's weather.
- Applying T to the state vector $(1, 0)$: tomorrow-sunny = $0.7(1) + 0.4(0) = 0.7$; tomorrow-rainy = $0.3(1) + 0.6(0) = 0.3$. So there is a 70% chance of sunny weather and a 30% chance of rain tomorrow.
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Diagonalization and powers of a matrix

- 9 Matrix M (rows (4, 1), (2, 3)) has eigenvalues $\lambda = 5$ and $\lambda = 2$ (found earlier), with eigenvectors $(1, 1)$ and $(1, -2)$ respectively. Explain, in general terms, why diagonalizing a matrix makes computing large powers like M^{10} much easier than repeated direct multiplication.
- If $M = PDP^{-1}$ where D is a diagonal matrix of eigenvalues, then $M^n = PD^nP^{-1}$. Raising a DIAGONAL matrix to a power is trivial — just raise each diagonal entry to that power — whereas repeatedly multiplying the full matrix M by itself n times is far more computationally intensive. This makes diagonalization extremely valuable for finding high powers of a matrix efficiently.
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Determinants of 3×3 matrices

- 10 Matrix N has rows (1, 2, 3), (0, 1, 4), and (5, 6, 0). Find $\det(N)$ by cofactor expansion along the first row.
- $$\det(N) = 1(1 \cdot 0 - 4 \cdot 6) - 2(0 \cdot 0 - 4 \cdot 5) + 3(0 \cdot 6 - 1 \cdot 5) = 1(-24) - 2(-20) + 3(-5) = -24 + 40 - 15 = 1.$$
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Identity matrix and matrix powers

- 11 Explain why $MI = IM = M$ for any square matrix M and the identity matrix I of matching size, and use this idea to find M^2 where M has rows (1, 2) and (0, 1).
- The identity matrix I (with 1s on the diagonal and 0s elsewhere) leaves any matrix unchanged under multiplication, playing the same role as the number 1 in ordinary multiplication — this is exactly why it is called the identity. $M^2 = M \times M$: row 1 = $(1(1) + 2(0), 1(2) + 2(1)) = (1, 4)$; row 2 = $(0(1) + 1(0), 0(2) + 1(1)) = (0, 1)$. So M^2 has rows (1, 4) and (0, 1).
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Using the inverse to solve a real-world system

- 12** A shop sells two products. Selling 2 units of product A and 3 units of product B brings in 76 dollars. Selling 4 units of A and 1 unit of B brings in 68 dollars. Write this as a matrix equation and solve for the price of each product using the coefficient matrix M with rows $(2, 3)$ and $(4, 1)$.

$\det(M) = 2(1) - 3(4) = 2 - 12 = -10$. $M^{-1} = \frac{1}{-10} \times$ the matrix with rows $(1, -3)$ and $(-4, 2)$, giving rows $(-0.1, 0.3)$ and $(0.4, -0.2)$. Applying M^{-1} to $(76, 68)^T$: price of $A = -0.1(76) + 0.3(68) = -7.6 + 20.4 = 12.8$; price of $B = 0.4(76) - 0.2(68) = 30.4 - 13.6 = 16.8$. So product A costs 12.80 dollars and product B costs 16.80 dollars.

Long-run behavior of a Markov chain

- 13** For the weather transition matrix T from earlier in this worksheet (rows $(0.7, 0.4)$ and $(0.3, 0.6)$), find the steady-state probability distribution $(p, 1 - p)$, which satisfies $T(p, 1 - p)^T = (p, 1 - p)^T$.

Setting up the steady-state equation using the first row: $0.7p + 0.4(1 - p) = p$. Expand: $0.7p + 0.4 - 0.4p = p$, so $0.3p + 0.4 = p$, giving $0.4 = 0.7p$, so $p = \frac{0.4}{0.7} = \frac{4}{7}$. The steady-state distribution is $(\frac{4}{7}, \frac{3}{7})$ — in the long run, the weather is sunny about $\frac{4}{7} \approx 57\%$ of the time, regardless of today's starting condition.

Singular matrices and when an inverse fails to exist

- 14** Matrix N has rows $(2, 4)$ and $(1, 2)$. Show that N has no inverse, and explain what this means about the corresponding system of linear equations.

$\det(N) = 2(2) - 4(1) = 4 - 4 = 0$. Since $\det(N) = 0$, N is singular and has no inverse. This means the corresponding system of linear equations (with N as the coefficient matrix) does not have a unique solution — it either has no solutions or infinitely many, depending on the constants on the right-hand side, since the two equations represented by the rows of N are scalar multiples of each other (row 2 is exactly half of row 1).

Transpose of a matrix and its properties

- 15** Matrix A has rows $(1, 2, 3)$ and $(4, 5, 6)$. Write down A^T (the transpose), and state its dimensions. The transpose is formed by turning rows into columns. A is a 2×3 matrix, so A^T is a 3×2 matrix with rows $(1, 4)$, $(2, 5)$, and $(3, 6)$.

- 16** Explain, using the general rule $(AB)^T = B^T A^T$, why the order of multiplication must be reversed when taking the transpose of a matrix product.

The rule $(AB)^T = B^T A^T$ (not $A^T B^T$) follows from how matrix multiplication combines rows and columns: transposing swaps rows and columns for both matrices, but for the product AB to remain dimensionally valid after transposing, the multiplication order must also reverse — if A is $m \times n$ and B is $n \times p$, then AB is $m \times p$ and $(AB)^T$ is $p \times m$; computing B^T ($p \times n$) times A^T ($n \times m$) correctly gives a $p \times m$ matrix, whereas $A^T B^T$ would generally not even be a valid multiplication.

- 17** Matrix A has rows $(2, 0)$ and $(0, 5)$. Find A^2 and A^3 , and describe the pattern for A^n in general.

Since A is diagonal, powers are found by raising each diagonal entry to that power. A^2 has rows $(4, 0)$ and $(0, 25)$. A^3 has rows $(8, 0)$ and $(0, 125)$. In general, A^n has rows $(2^n, 0)$ and $(0, 5^n)$ — for a diagonal matrix, raising it to a power simply raises each diagonal entry to that power, since off-diagonal zero entries stay zero throughout the multiplication.